AgroGuard 🌿

We built a device that protects farmers' crops — automatically, without internet, without chemicals.

💧 Waters the farm itself

🐛 Spots pests by camera

📱 Shows farmer what's happening

Meet Musa Caleb.

A farmer in Jos North. He grows maize and yam to feed His family of seven.

₦180,000	₦60,000	₦120,000	₦600,000
What He should earn	What He actually earns	Money lost every season	Lost over 5 years (a house. school fees. a future.)

🌼 He waters by memory.

Too much water — His yam roots rot. Too little — His maize dries up and dies. He has no way to know when to water.

🐛 The pests arrive quietly.

Armyworms eat through His maize overnight. By the time He sees the damage, half His crop is gone. He cannot afford to lose this.

💸 The loss never stops.

He borrows money to survive. Next season, He does everything the same way — because nobody showed Him a better way.

How Bad Is The Problem?

4 in 10
crops on Jos Plateau are destroyed by pests every year

3 in 10
crops die because farmers watered too much or too little

9 in 10
local farmers have small farms with no technology

ZERO
affordable farm devices available in Jos today

What happens when a farmer loses their harvest?

Crops fail → No money → Can't pay school fees → Children go hungry → Family leaves farming → Food gets scarce for everyone

Why the old ways do not work

Chemical spray	Very expensive · makes children sick · harms the soil
Farm officers	One officer covers 200+ farms · most farmers never see them
Guessing	No data · no warning · damage found only when it is too late
Imported devices	Cost ₦500,000+ · need internet · nobody knows how to fix them

What If Musa Caleb Had AgroGuard?

❌ Without AgroGuard

He waters when He remembers — crops die from too much or too little

He finds pests only after half the crop is already destroyed

He buys expensive chemical spray that hurts His soil and His family

He has no record of what happened or why

Every season, He loses money and wonders what went wrong

✅ With AgroGuard

The device waters the farm automatically when soil is dry

The camera spots pests in less than 2 seconds — a buzzer chases them away

No chemicals needed for pest detection — saves thousands of naira

His phone shows a log of everything that happened on the farm

More crops survive. More money for school fees and food.

"I water when I remember. I spray when I see damage. By then it is too late." — Musa Caleb, Jos North

LIVE DEMO

Watch AgroGuard work right now.

- ✓ Camera detects a pest — buzzer fires in under 2 seconds
- ✓ Soil sensor reads moisture level live
- ✓ Dashboard on phone shows everything happening on the farm

We built this. We tested this. This is real.

How AgroGuard Works

Part 1: Automatic Watering

A small sensor sits in the soil.
Every 5 seconds it checks: is the soil dry?
If dry → water pump turns ON automatically.
When wet enough → pump turns OFF.

Part 2: Pest Detection

A camera watches the plants all the time.
An AI looks at every picture it takes.
If it sees a pest → buzzer sounds immediately.
This happens in less than 2 seconds.

↓ *Both parts talk to a small computer (Raspberry Pi) which runs everything* ↓

Part 3: The Farmer's Dashboard

Open any browser on any phone. See: how wet is the soil? Was a pest spotted? Is the pump running? Everything in one place, in real time. No internet needed — it works on local Wi-Fi.

What we used to build it

Arduino board	Soil moisture sensor	Raspberry Pi mini-computer	Camera	Buzzer
controls the water pump	checks if soil is dry	runs the AI and dashboard	watches for pests	sounds the alarm on pests

5 Pests AgroGuard Can Spot

Aphid

- Attacks: Tomato, pepper, cabbage
- Sign: Leaves curl and turn yellow

Armyworm

- Attacks: Maize, millet, sorghum
- Sign: Leaves eaten overnight

Mealybug

- Attacks: Yam, cassava, citrus
- Sign: White powder on stems

Stem Borer

- Attacks: Maize, rice, sugarcane
- Sign: Holes in stem, plant falls over

Weevil

- Attacks: Stored beans and grains
- Sign: Tiny holes in stored food

Our AI Model

- ✓ Trained on hundreds of real pest photos
- ✓ Correct 9 out of 10 times
- ✓ Gives answer in under 2 seconds
- ✓ Works without internet

What This Means for Farmers

3 in 10

less crops wasted
through smarter watering

4 in 10

fewer pests destroying
farms if caught early

₦50,000+

saved per farmer
by not buying chemical spray

2 seconds

from spotting a pest
to sounding the alarm

For Farmers

- More crops survive each season
- No need to guess when to water
- Pests found before they spread
- Dashboard shows what is happening even at night
- No expensive chemicals for detection

For Jos Plateau

- More food available in local markets
- Families earn more from their farms
- Children stay in school — school fees get paid
- Young people learn how to build tech for their own community
- A model other states

can copy

💡 Why AgroGuard Is Different

We did not invent AI. We did not invent irrigation.

We made it work for a Jos Plateau farmer's pocket, phone, and reality.

Option	Cost	Needs internet?	Knows local pests?	Can be fixed locally?
Chemical spray	₦3,000–8,000/month	No ✅	No ❌	No ❌
App on phone (Plantix)	Free but needs internet	Yes ❌	No ❌	No ❌
Imported smart device	₦500,000+	Yes ❌	No ❌	No ❌
AgroGuard ✅	Affordable	No ✅	Yes ✅	Yes ✅

"AgroGuard is not new technology. It is the first time this technology was built for a Jos farmer's budget and offline reality."

🗺️ Now Next Future = SCALABILITY

✅ NOW — What We Have Built

- A working device that waters the farm automatically
- A camera that spots 5 types of pests
- A buzzer that chases pests away the moment they appear
- A dashboard a farmer can open on any phone
- All of this works without internet

⬜ NEXT — Our Next Steps

- Test on a real farm in Jos with actual farmers
- Add a dedicated camera for better night vision
- Send text alerts to farmer's phone when a pest is found
- Build a wet-proof cover for outdoor use
- Collect feedback from 10 real farmers

🚀 FUTURE — Where We Want To Go

- Protect farms across all of Plateau State
- Detect crop diseases (not just pests)
- Let farmers share data with each otHis
- Solar powered — works with no electricity
- Share the code free with other students in Nigeria

🤝 We Are Not Asking for Money

We are asking for partners who believe farmers deserve better tools.

🌾 Farmers who want to test it

We need 10 real farmers in Jos North who will let us install AgroGuard on their farm and give us honest feedback on what works and what does not.

🏫 Teachers and schools

We want to run workshops where secondary school students learn how to build and repair this device themselves — using materials sold right here in Jos.

🛠️ Mentors who know farming

We are students. We built the technology. We need experienced farmers and community leaders to help us understand real farm needs better.

📡 Help spreading the word

If you know a farmer, an NGO, a mosque, a church, or a community group — tell them about AgroGuard. Every connection helps a farmer get protected sooner.

We did not build this to impress judges. We built it to help Musa Caleb.

AgroGuard

AI for the field. Not for the cloud.

"We built AI that works offline, in the field, protecting the crops that feed families."

✅ **We tested it**	Over 700 real camera captures. Working demo running right now.
🌾 **We built it for Musa Caleb**	Not for Silicon Valley. For Jos Plateau farmers who have no alternative.
📖 **The code is open**	github.com/Muwatta/agroguard — any student can learn from it.

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